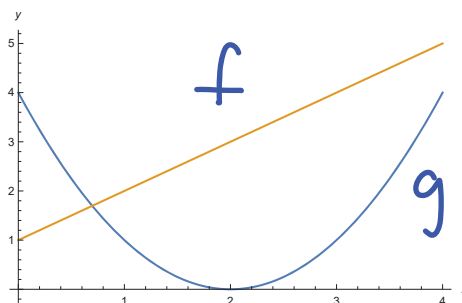


SECTION A [Multiple choice questions] - 15 marks

QUESTION 1

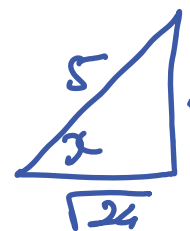
Consider the graphs of a straight line f and a parabola g below.



$$g(f(1)) = g(2) = 0$$

$$(g \circ f)(1) =$$

☒ (a) 0	(b) 1	(c) 2	(d) 3	(e) 4	(f) 5	(g) None of (a) - (f)
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QUESTION 2

If $\sin x = -\frac{1}{5}$, $\frac{3\pi}{2} < x < 2\pi$, then $\sin(2x) = 2\sin x \cos x = 2 \cdot \frac{-1}{5} \cdot \frac{\sqrt{24}}{5} = \frac{-2\sqrt{24}}{25}$

(a) $-\frac{2}{5}$	(b) $-\frac{1}{10}$	☒ (c) $-\frac{4\sqrt{6}}{25}$	(d) $-\frac{\sqrt{24}}{25}$	(e) $\frac{4\sqrt{6}}{25}$	(f) $-\frac{8}{25}$	(g) None of (a) - (f)
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QUESTION 3

$$b = a^3 + 3$$

$$a = 0: b = 3$$

$$a = 7: b = 52$$

The relationship between the variables a and b is given by $b = a^3 + 3$. Which b values correspond with the a values in the interval $[-1, 7]$?

(a) $b \in [-1, 52]$	(b) $b \in [1, 52]$	☒ (c) $b \in [3, 52]$	(d) $b \in [4, 52]$	(e) None of (a) - (d)
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QUESTION 4

If $f(x) = \ln \sqrt{x-1}$, then $f^{-1}(x) =$

$$y = \ln \sqrt{x-1}. \text{ Let } x = \ln \sqrt{y-1} \Rightarrow e^x = e^{\ln \sqrt{y-1}} = \sqrt{y-1}$$

(a) $e^{\frac{x}{2}} + 1$	(b) $e^{\frac{x}{2}+1}$	☒ (c) $e^{2x} + 1$	(d) e^{2x+1}	(e) $e^{\sqrt{x-1}}$	(f) $\frac{1}{2}e^{x-1}$	(g) None of (a) - (f)
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QUESTION 5

For which value(s) of x is $y = \cos^{-1}(3 + 2x) = \arccos(3 + 2x)$ defined?

$$\cos y = 2x + 3 \Rightarrow -1 \leq 2x + 3 \leq 1 \Rightarrow -4 \leq 2x \leq -2$$

(a) $x \in [-1, 1]$	☒ (b) $x \in [-2, -1]$	(c) $x \in [0, \pi]$	(d) $x \in [-3, -\frac{1}{2}]$	(e) $x \in (-1, 1)$	(f) $x \in \mathbb{R}$
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QUESTION 6

The average rate of change of $f(t) = t^2 + 2$ over the interval $[2, 4]$ is

$$\frac{f(4) - f(2)}{2}$$

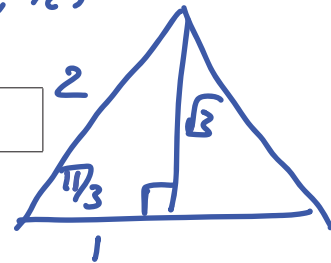
(a) 2	(b) 3	(c) 4	(d) 5	☒ (e) 6	(f) None of (a) to (e)
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QUESTION 7

The value of $\tan^{-1}(-\sqrt{3}) = \arctan(-\sqrt{3})$ is:

(a) $-\frac{\pi}{6}$	(b) $\frac{11\pi}{6}$	(c) $\frac{5\pi}{6}$	(d) $\frac{2\pi}{3}$	(e) $\frac{5\pi}{3}$	☒ $-\frac{\pi}{3}$	(g) None of (a) - (f)
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$$\tan y = -\sqrt{3} \text{ and } y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$



QUESTION 8

The solution of $\ln(\ln(3x)) = 1$ is

(a) $x = 0$	(b) $x = \frac{1}{3}$	(c) $x = \frac{1}{3}e$	☒ $x = \frac{1}{3}e^e$	(e) $3e$	(f) None of (a) to (e)
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$$e^{\ln(\ln(3x))} = e^1 \Rightarrow \ln(3x) = e$$

$$3x = e^e$$

QUESTION 9

Given: $y = \ln(-x)$. The equation of the graph that results from shifting two units to the right is:

(a) $y = \ln(x) + 2$	☒ $y = \ln(-x + 2)$	(c) $y = \ln(-x - 2)$	(d) $y = \ln(x - 2)$	(e) $y = \ln(x + 2)$	(f) None of (a) - (e)
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QUESTION 10

Consider the three statements:

(1) $\sec(x) = -500$

(2) $x^2 + x = 1$

(3) $|x + 1| < -1$

$$\cos x = \frac{-1}{500} \checkmark$$

$$x^2 + x - 1 = 0 \Rightarrow b^2 - 4ac$$

$$= 1 - 4(1)(-1)$$

$$= 5 > 0$$

Which of these has (have) no solution?

(a) Only (1).

(b) Only (2).

☒ (c) Only (3).

(d) Only (1) and (2).

(e) Only (1) and (3).

(f) Only (2) and (3).

(g) (1), (2) and (3).

(h) Statements (1), (2) and (3) have solutions.